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(py)ModRev – Repairing inconsistent Boolean logical models of biological regulatory networks

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

BioLogics Workshop

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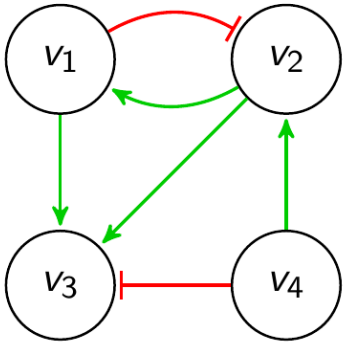
Boolean logical model of regulatory networks

A regulatory network is a collection of molecular compounds that interact with each other.

Boolean logical model:

- Compounds represented by a Boolean variable
- Interaction defined as positive (activation) or negative (inhibition)
- Regulations defined as Boolean functions



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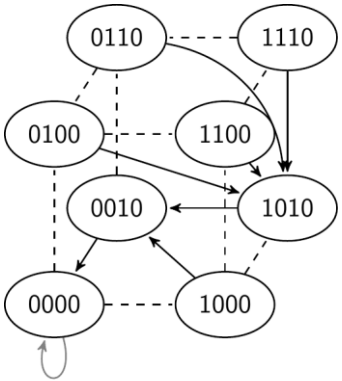
Dynamics

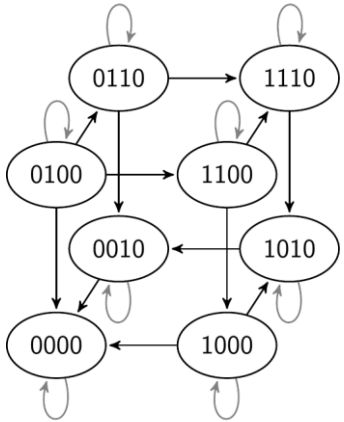
The value of each node can change in time defining the state of the network.

The regulatory functions update the value of the corresponding node.

Different update schemes can be considered, defining State Transition Graphs.

Synchronous





Asynchronous

Model Revision

As new experimental data becomes available, models may become inconsistent!

- Not being able to reproduce the new information.
- Models need to be revised.

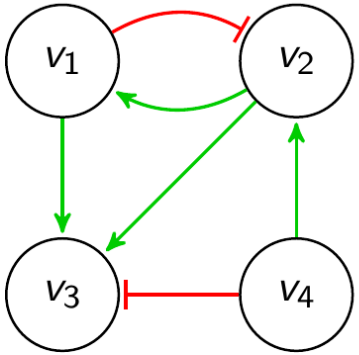
A model is consistent if all its nodes are consistent.

- Value of each node given by its regulatory function is equal to the observed value.

A model is inconsistent otherwise.

How can we repair an inconsistent model?

- Change a regulatory function?
  - 2^{2^k} possibilities for each node!
- Change the type of interactions?
- Add or remove interactions?



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There are $\approx 10^{24}$ possible combinations!

(65536 Boolean functions with 4 regulators)

Repair Operations

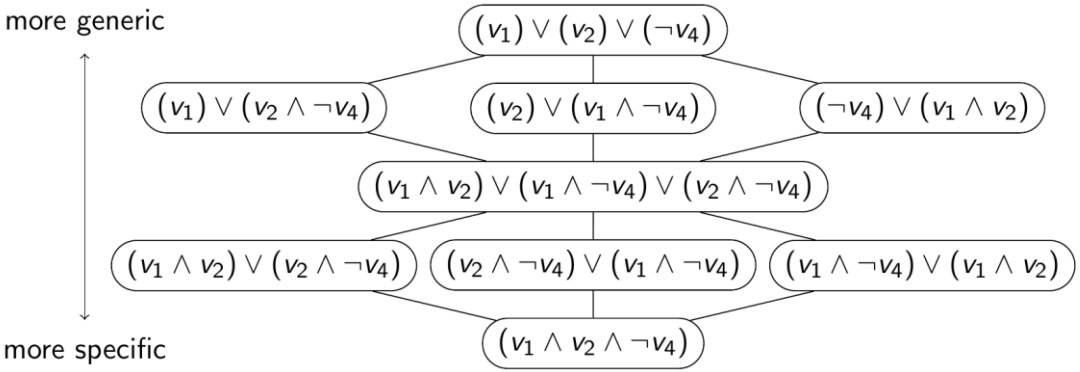
Cause	Repair Operation
Wrong regulatory function	Function change
Wrong interaction type	Edge sign flip
Wrong regulator	Remove edge
Missing regulator	Add edge

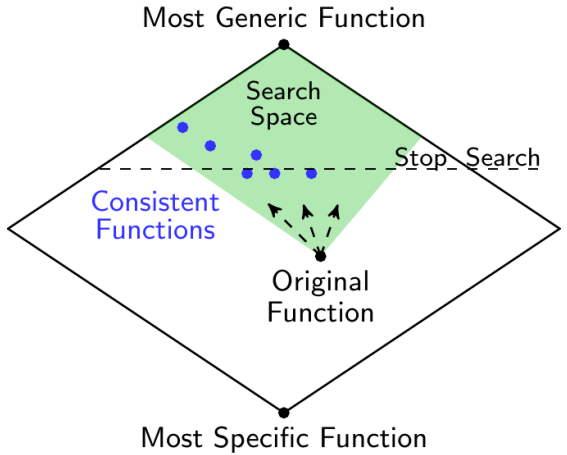
Optimization Criteria:

1. Minimize interaction addition/removal.
2. Minimize interaction type changes.
3. Minimize Boolean function changes.

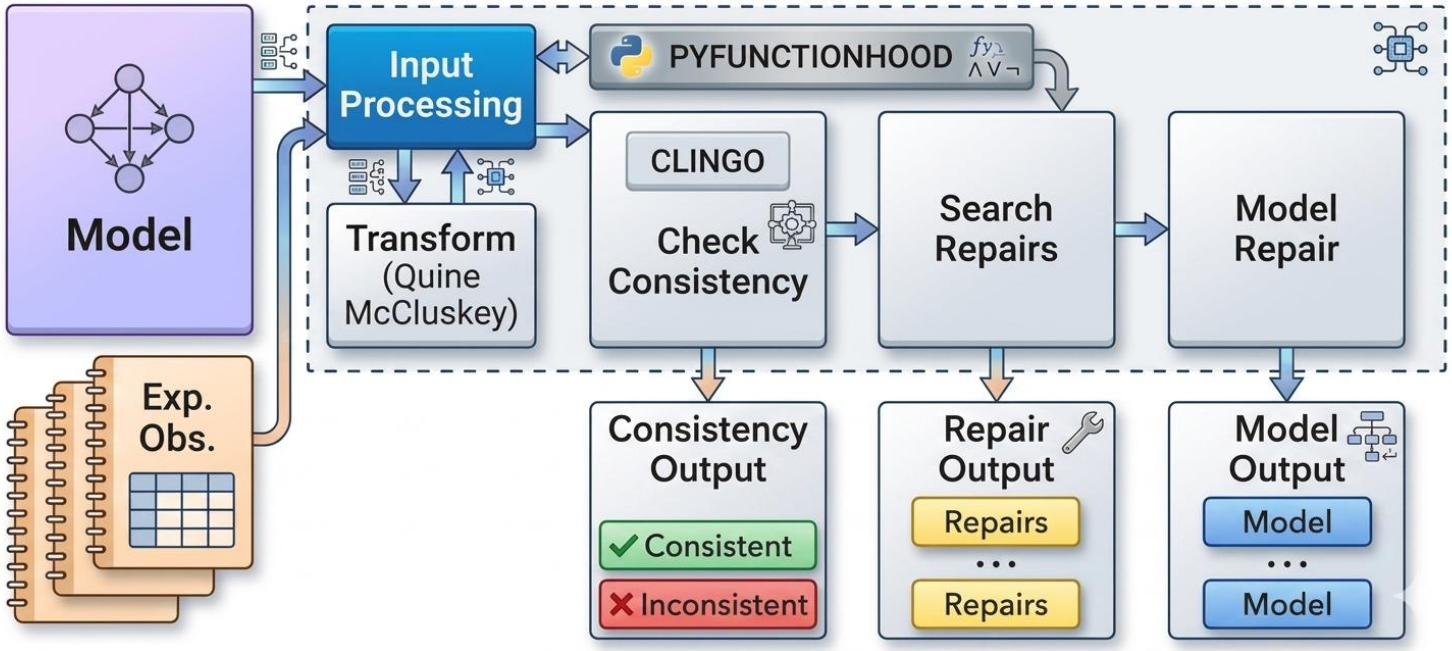
Function Repair

We consider monotone non-degenerate Boolean functions.

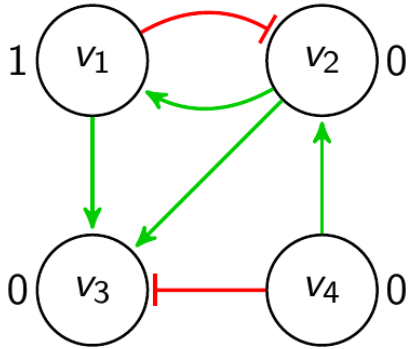




Overview


Model
Exp.
Obs.
Input
Processing
PYFUNCTIONHOOD
Transform
(Quine
McCluskey)
CLINGO
Check
Consistency
Search
Repairs
Model
Repair
Consistency
Output
Consistent
Inconsistent
Repair
Output
Repairs
Repairs
Model
Output
Model
Model

Model Repair Example (Stable State)



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- Consistent model!

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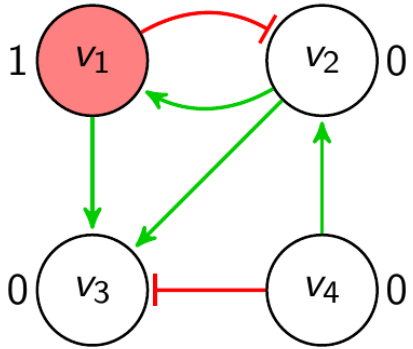

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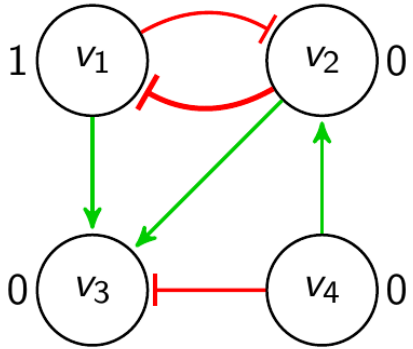

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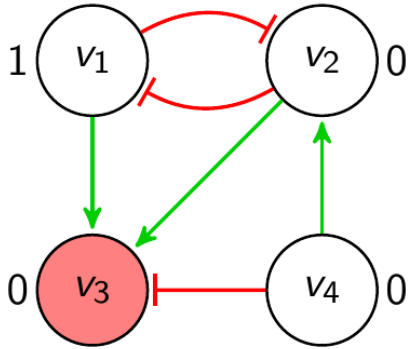

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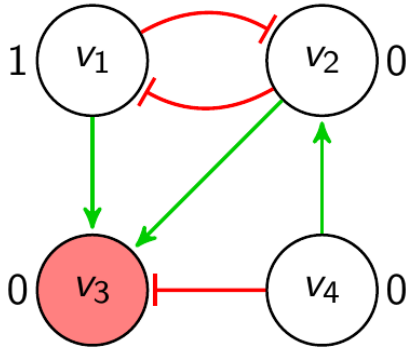

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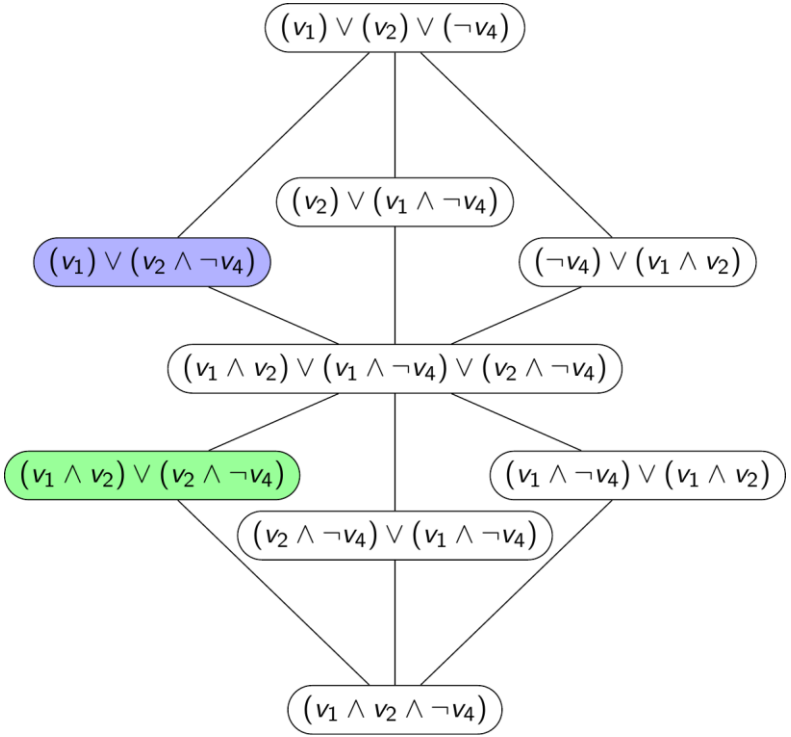

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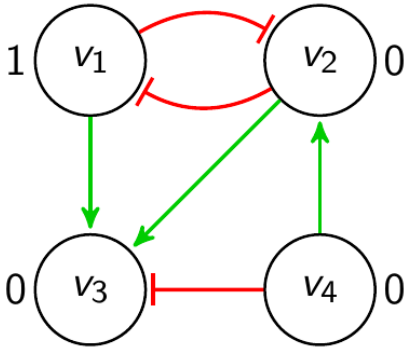

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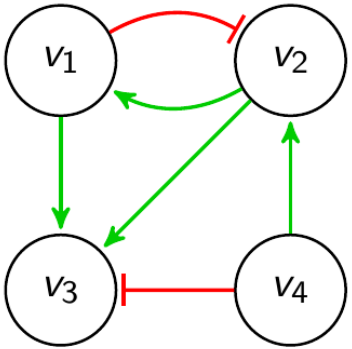
Model Repair Example (Time-series synchronous)



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Time

Node	t_0	t_1	t_2
v_1	1	1	0
v_2	1	0	1
v_3	0	1	0
v_4	1	1	1

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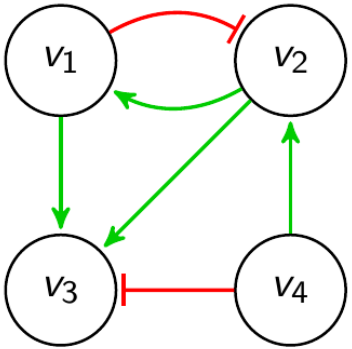

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ModRev and pyModRev – What can we do?

- Confront models with different experimental observations simultaneously.
  - Stable state and time-series.
- Different update schemes (simultaneously):
  - Synchronous, asynchronous and complete.
- Support missing values in observations.
  - Minimize number of inconsistent nodes.
- Avoid given state as steady state.
- Produce optimal repairs.

ModRev (C++)

pyModRev (python)

Next Steps

- Support different update schemes.
  - Most Permissive
- Full support to reachability properties.
  - Go from fixed to variable time steps
  - Check reachability
  - Avoid reaching some state
- Improve function repair search efficiency.

Thank you!

pyModRev

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